

Stewart Slocum

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Education

Ph.D., *MIT CSAIL*

August 2022 - March 2025 (On Long-Term Leave)

B.S. CS and Applied Math, *Johns Hopkins University*

August 2017 - May 2021

Experience

xAI *Member of Technical Staff*

October 2025 -

Technical lead for team of 3 on model behavior and safety training for Grok 4.2 and Grok 4.3:

- **Model behavior training:** Led alignment and model behavior training for Grok 4.2. Reduced behavioral issues from Grok 4.1 (sycophancy, overconfidence, tool call fabrication) while preserving high user preference scores (Grok 4.2 #3 on LMSYS chat arena at release). Led chat training for Grok 4.3 (RLHF, writing quality, refusals, instruction-following).
- **Non-verifiable reasoning:** Developed a new post-training method that improved writing quality and reasoning ability on non-verifiable domains. Increased winrate on internal user preference leaderboard by 10%.
- **Alignment and safety evaluations:** Built xAI's first holistic pre-deployment alignment audit, led [Grok 4.1](#) and [Grok 4.2](#) model cards, and ran third-party testing with nonprofit evaluators (METR, Apollo) and government bodies (UK AISI, US CAISI). Performed post-deployment patching on several urgent safety issues.

Anthropic *Safety Research Fellow*

April 2025 - October 2025

Projects on synthetic document finetuning and interpretability:

- [Believe It or Not: How Deeply do LLMs Believe Implanted Facts?](#): Built methods to deeply implant false facts into LLMs and measure how genuine the implanted beliefs are, through behavioral and model internals testing. These methods have since been used for alignment mid-training and research at several labs and alignment research organizations. Discovered several counterintuitive results that led to follow-up papers on finetuning science and model diffing. Supervised by Samuel Marks and Ryan Greenblatt.
- [Narrow Finetuning Leaves Clearly Readable Traces in Activation Differences](#): Ran experiments to reverse-engineer the content of finetuning datasets through model diffing interpretability techniques. Used these technique to understand some mechanistic effects of narrow finetuning on LLMs. Supervised by Neel Nanda. ICLR 2026.

Algorithmic Alignment Group, MIT *PhD Student*

August 2022 - March 2025

Technical tools for AI alignment and safety. A few projects:

- [Diverse Preference Learning for Capabilities and Alignment](#): method inspired by social choice theory to mathematically understand and address mode collapse in aligned LLMs. This method aligns output distributions to population-level preferences, improving diversity-quality tradeoffs, social representation, and best-of-N inference scaling. ICLR 2025.
- [Model Tampering Attacks Enable More Rigorous Evaluations of LLM Capabilities](#): used fine-tuning and latent-space attacks to estimate and upper-bound vulnerabilities from input-space adversaries. TMLR 2025.
- [Beyond Binary Rewards: Training LMs to Reason about Their Uncertainty](#): RL technique for LLMs that leads to emergent reasoning about their uncertainty. This improves calibration while maintaining performance and allows for novel test-time scaling procedures. ICLR 2026.